

Subject Description Form

Subject Code	AAE5205
Subject Title	Aircraft Engine Systems and Combustion
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To provide students with fundamental knowledge of modern aircraft gas-turbine engines, covering advanced aircraft engine systems and the theoretical principles of combustion science applied within them.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Possess a fundamental understanding of state-of-the-art aircraft gas turbine engines, focusing more on subsystems of aircraft gas turbine engines; - Possess a deep understanding of combustion theories and effectively apply knowledge to investigate various combustion-related problems.
Subject Synopsis/ Indicative Syllabus	Introduction to propulsion – Brief history of propulsion, aircraft piston engines, aircraft gas turbine engines, performance parameters of turbofan engines, other air-breathing engines; Aircraft engine subsystems – Nacelle, inlet, nozzle, fan, compressor, turbine, combustor, afterburner, fluid system (air, oil, and fuel), control system, engine test and maintenance, engine start; Introduction to combustion – Brief history of combustion science, combustion modes, and flame types Thermochemistry – Review of thermodynamics, stoichiometry, chemical equilibrium, adiabatic flame temperatures; Mass transfer – Stefan problem, droplet evaporation and burning; Chemical kinetics – Global/elementary reactions, rates of reaction; Mass transfer – Stefan problem, droplet evaporation; Chemical kinetics – Global/elementary reactions, rates of reaction, some important chemical mechanisms, four types of reactors, simplified conservation equations for reacting flows; Flames – Laminar premixed flame, laminar diffusion flame, turbulent premixed flame, turbulent diffusion flame Fuels and low-carbon combustion – Hydrogen/ammonia, combustion instabilities

Teaching/Learning Methodology	1. The teaching and learning methods include lectures, homework assignments, lab reports, and a final examination. 2. The continuous assessment and examination are aimed at providing students with the integrated knowledge required for propulsion systems. 3. Technical/practical examples and problems are raised and discussed in class. 4. Experiments projects are designed to understand the combustion system. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="2">Intended subject learning outcomes to be covered</th></tr><tr><th>a</th><th>b</th></tr><tr><td>1. Lectures</td><td>✓</td><td>✓</td></tr><tr><td>2. Homework assignments</td><td>✓</td><td>✓</td></tr><tr><td>3. Experiments/Projects</td><td>✓</td><td>✓</td></tr><tr><td>4. Final exam</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes to be covered		a	b	1. Lectures	✓	✓	2. Homework assignments	✓	✓	3. Experiments/Projects	✓	✓	4. Final exam	✓	✓					
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="2">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th></tr><tr><td>1. Homework</td><td>20%</td><td>✓</td><td>✓</td></tr><tr><td>2. Lab session</td><td>30%</td><td>✓</td><td>✓</td></tr><tr><td>3. Final examination</td><td>50%</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="2"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Final Examination}$ The continuous assessment consists of homework and one lab report. They are aimed at evaluating the progress of students study, assisting them in self-monitoring of fulfilling the respective subject learning outcomes, and enhancing the integration of the knowledge learned. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		a	b	1. Homework	20%	✓	✓	2. Lab session	30%	✓	✓	3. Final examination	50%	✓	✓	Total	100%		
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Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Lab session	6 Hrs.
	Other student study efforts:	
	▪ Self Study	39 Hrs.
	▪ Lab preparation and homework	39 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Elements of Propulsion: Gas Turbine and Rockets, 2nd Edition, 2006. Jack Mattingl., AIAA. 2. The Jet Engine, 5th Edition, Rolls Royce, Wiley 3. An Introduction to Combustion: Concepts and Applications, 4th Edition, 2021. Turns, S. et al., McGraw Hill. 4. A Gallery of Combustion and Fire, 1st Edition, 2020. Agarwal, A. et al., Cambridge University.	

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